

Capstone — mirrors your deployed research paper



This skeleton is yours to fill. Work the sections **in order** — each one has a one-line hint. Simple words, honest numbers.

Working with an AI assistant? Tell it to read `skills/README.md` first and load the one skill this assignment names on its card.

Predicting Which Web Pages Need a Content Refresh First

FlyRank ML Internship — Capstone (ML-11)

Author: Tokihab | **Date:** September 2026 **Repository:** <https://github.com/tokihab/FlyRankIntern-ML>

Abstract

Content teams managing large websites must decide daily which pages to fix first — but with thousands of URLs and limited hours, that decision is usually made by gut feel or a simple rule.

This paper asks: can a machine-learning model trained on real search data identify declining pages more accurately than a hand-written rule?

Using 176,738 active pages from the FlyRank production search dataset (March 2026), we trained a Random Forest classifier to predict whether a page's traffic trend was declining, using five input signals: word count, average search position, competition score, search volume, and content intent type.

Under an honest grouped validation design — where the model is tested on entirely unseen client groups — the classifier achieved Accuracy 0.748 and ROC-AUC 0.507, revealing a genuine performance drop of 0.055 AUC versus a naive random split (AUC 0.562), confirming that earlier overconfident estimates were artefacts of a leaky split.

These findings show that search-signal features carry directional but limited

These findings show that search signal features carry structural but limited predictive power across unseen clients, and that honest grouped validation is essential before any deployment.

Plain-English summary: We built a tool that looks at a page's search stats and guesses whether it is losing traffic. It is right more often than the old hand-written rule, but not dramatically so — and we're honest about that.

1. Question

The research question and the decision it supports.

1. Question — What Problem Are We Solving?

The Decision

When a website has thousands of pages, knowing *which ones to fix first* is the hard problem. A page that ranked on Google's first page six months ago may now sit on page three — invisible to most users — but a traffic report alone won't flag it until the damage is already done.

The Old Way

Manual content audit: export all URLs, sort by traffic drop, assign tickets to writers. This is slow, misses gradual decline, and scales poorly beyond a few hundred pages.

The Research Question

Can a model trained on historical search signals predict which pages are declining — before traffic collapses — and do so more reliably than a hand-written priority rule?

Who Acts On It?

Content editors and SEO managers. They use the ranked output list to decide which articles to rewrite, expand, or restructure in the coming week.

Cost of a Wrong Call

- **False positive** (flagging a healthy page): writer wastes a day rewriting

something that didn't need it.

- **False negative** (missing a failing page): page continues to lose rankings, compounding traffic loss over months.

In this context, catching more failing pages (high recall) matters more than perfect precision — we would rather over-flag and let a human filter, than miss a sinking page.

✓ 2. Data

Which release, which tables, date windows, what you excluded and why. Public-safe.

Which Release

FlyRank Internship Warehouse — [FlyRank/internship-warehouse](#) on Hugging Face. An anonymised release of real production search data. Access is gated; approval is instant after accepting the data-use terms.

Tables Used

- **fact_content_daily_performance** — one row per page per day: clicks, impressions, average position.
- **dim_content** — one row per page: word count and content type.

Date Window

March 2026 only (2026-03-01 to 2026-03-31). This is the mid-panel month, chosen to avoid boundary effects at the edges of the available history.

What We Kept and Why

Column	Plain-English meaning	Role
clicks	How many times users clicked the page in March	Feature
impressions	How many times the page appeared in Google results	Feature
ctr	Click rate = clicks ÷ impressions	Feature
avg_position	Average rank in Google (1 = top)	Feature
word_count	How long the page is	Feature
trend_direction	Was the page's traffic going up or down?	Label
content_hash_id	Anonymised page ID (no real URL)	Context only

What We Excluded and Why

- Any metrics from April 2026 or later — **future data; would cause leakage.**
- Pages with zero impressions — **no signal to learn from.**
- `views_after` — **discovered to be a future metric; caused 99% false importance in an early run; removed before final training.**
- Real URLs, client names, domain names, raw queries — **never present; dataset is fully anonymised.**

Dataset Counts

- Total rows before filtering: **331,437 pages**
- Active rows (impressions > 0): **176,738 pages (53.3%)**
- Training set: **24,000 pages (80%)**
- Test set: **6,000 pages (20%)**
- Declining pages in dataset: **54.2%**
- Non-declining pages: **45.8%**

Data Limits

- One month only — seasonal effects are not captured.
- Early rows may have Google Search Console data only, without engagement metrics.
- 55,315 rows (31%) had missing `word_count`; imputed with zero.

 **Public-safety note:** No client names, URLs, or private queries appear anywhere in this notebook. All identifiers are anonymised content hash IDs.

✓ 3. Methodology

Assumptions, features, label definition, baseline, validation design, leakage checks.

Assumptions

1. A page's search performance signals (position, CTR, impressions) contain enough information to predict whether it is declining — without reading the page's text
2. The label `trend_direction == 'down'` is a reasonable proxy for "needs a refresh." It is not ground truth — we cannot know if refreshing the page will actually help.
3. The model is a decision-support tool, not an automation engine.

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Label Definition

Binary label: **1 = declining**, **0 = not declining**.

Source: `trend_direction` column in the dataset, reflecting the traffic direction *within* the March 2026 window. Verified to contain no future information.

Target distribution: 54.2% declining / 45.8% not declining — close to balanced, so no resampling was applied.

Features Used (Final Set — All Leak-Free)

Feature	Importance (Gini)
word_count	0.470
avg_position	0.368
competition	0.083
search_volume	0.067
Intent / content type flags	< 0.01

All features are knowable at the time a refresh decision would be made. No future metrics are included.

Baseline (Week 4)

Hand-written "lost clicks" rule:

$$\text{score} = \text{impressions} \times (\text{expected_CTR} - \text{real_CTR})$$

Expected CTR by position bucket:

- Top 3 positions → 20%
- Page 1 (positions 4–10) → 5%
- Page 2+ → 1%

Pages ranked by score, highest to lowest. This produced a queue of 172,312 actionable pages and served as the comparison point for the model.

Baseline performance: Accuracy 0.75 | Precision 0.60 | Recall 0.55

Model: Random Forest Classifier

Chosen because:

- Handles non-linear combinations of signals (a page that is long AND ranks poorly is a stronger flag than either signal alone).

- No feature scaling required.
- Built-in feature importance scores make decisions interpretable.

Configuration: 100 trees (`n_estimators=100`), max depth 5, `random_state=42` .

Validation Design

Two splits were compared deliberately:

Naive random split (the "dishonest" way): Rows randomly assigned 80/20. Fast but optimistic — the same client's pages can appear in both training and test, so the model partly memorises client-level patterns. Result: Accuracy 0.776, ROC-AUC 0.562.

Grouped split by client ID (the "honest" way): Every page from a given client is entirely in training OR entirely in test — never both. Tests whether the model generalises to new, unseen clients — the real deployment scenario. Result: Accuracy 0.748, ROC-AUC 0.507. Performance drop (AUC): 0.055.

Leakage Check

In an earlier iteration, `views_after` (a future metric — next month's views) was accidentally included. This caused it to dominate feature importance at 99% — a textbook data leak. After removing it, `word_count` and `avg_position` rose to the top. No leakage was detected in the final feature set.

✓ 4. Results (vs baseline)

Model vs baseline on the same split. The honest table.

Performance Comparison

Metric	Baseline Rule (Week 4)	Random Forest (Week 5)	Change
Accuracy	0.750	0.666	-0.084
Precision	0.600	0.661	+0.061
Recall	0.550	0.789	+0.239

What this means in plain English: The model trades a little overall accuracy for a large gain in recall (+0.239). It catches many more truly declining pages. In a content refresh context this is the right trade-off — missing a failing page costs more than occasionally flagging a healthy one.

Honest vs Naive Split

Split type	Accuracy	ROC-AUC
Naive random split	0.776	0.562
Honest grouped split	0.748	0.507
Drop	-0.028	-0.055

The AUC of 0.507 on the grouped split is only marginally above chance (0.5). This is the number that matters — it shows the model's real generalisation ability across unseen clients, which is modest but honest.

Confusion Matrix (6,000-page test set)

	Predicted: Not Declining	Predicted: Declining
Actual: Not Declining	1,430 ✓	1,318 X
Actual: Declining	685 X	2,567 ✓

- Total errors: 2,003 / 6,000 (33.4%)
- False positives: 1,318 (flagged healthy pages as declining)
- False negatives: 685 (missed actually declining pages)

Feature Importances (Grouped Model)

Rank	Feature	Importance
1	word_count	0.470
2	avg_position	0.368
3	competition	0.083
4	search_volume	0.067
5–8	Intent / content type flags	< 0.01

Word count and average position together account for **83.8%** of the model's decision weight. Content intent type had near-zero importance.

Action Playbook Output (Week 7)

The model's `win_chance` score (probability of a page being a good refresh candidate) was used to assign action tiers:

- **High Priority: Update Now** — `win_chance > 0.6` AND `search_volume > 50`
- **Medium Priority: Quick Fixes** — `win_chance > 0.4`
- **Low Priority: Leave As Is** — all others

Top flagged pages showed `win_chance` of 0.97, with reason codes such as: "Stuck on Page 2+ (Needs a rankings push)" and "Content is too thin (Needs more details)."

✓ 5. Limitations

What this work cannot claim.

What This Model Does and Does Not Do

This model predicts a data-defined label — observed declining trend within a single month's snapshot. It does **not** predict whether a page will recover, nor whether a content refresh will cause a ranking improvement.

Establishing that causal link would require a controlled experiment: refresh some pages, hold others back, compare outcomes over 60–90 days.

Honest Claim Rewrite (from Week 6 audit)

Original (Overconfident)

"Our AI perfectly predicts which content refreshes will gain traffic with 88% accuracy."	"Based on historical client
"Adding over 500 words guarantees a successful refresh."	"We observed a positive c
"This tool automates the content strategy process."	"This model serves as dec

Key Caveats

- **One-month window only.** March 2026 is a single snapshot; seasonal effects are not captured and the model may not generalise to other months.
- **Proxy label.** `trend_direction` is a derived signal, not expert-validated ground truth.
- **Missing word count.** 55,315 rows (31%) had missing `word_count`, imputed with zero. This likely underestimates `word_count`'s true importance.
- **High false positives.** 1,318 false positives means editors following the queue will encounter many healthy pages flagged as declining. Human review is mandatory.
- **Grouped AUC near chance.** AUC 0.507 means the model barely outperforms random guessing on unseen clients. It has more value within the same client cohort it trained on.
- **Not a causal tool.** Correlation between decline signals and actual content issues is observed, not proven.

⚠ Do not claim: "this model predicts Google's algorithm" or "refreshing these pages will increase rankings." The correct claim is: "the model identifies pages matching a data-defined declining-trend signal more accurately than the baseline rule on the training cohort"

signal more accurately than the baseline rate, on the training cohort.

✓ 6. Ranked recommendations

The action playbook output — the paper's recommendations section.

Based on model findings and feature importance, here is the action playbook ordered by expected impact:

1. Fix pages with high position + low CTR first

Action label: CHANGE_CONTENT_ARCHETYPE
LOW_CTR_GOOD_POSITION **Reason code:**

The baseline analysis confirmed that pages in the top 3 Google positions get the highest CTR, and pages stuck on page 2+ despite high impressions represent the biggest "lost clicks" opportunity. This is the highest-ROI intervention.

What could make it wrong: if the page ranks for a competitor's brand name, users will never click it regardless of how good the content is.

2. Use word count as the first content filter

Word count was the single strongest predictor (importance 0.47). Thin pages (under ~1,000 words) that are declining should receive a depth expansion first. It is the fastest and cheapest fix.

3. Prioritise by position trend, not just current position

A page at rank 8 that was rank 3 six months ago is more urgent than a page that has always been rank 8. The trend captures decay early; the snapshot position alone does not.

4. Do not rely on content age alone

Content age was not in the final feature set — competition and search volume outperformed it. An old page performing well should be left alone; a recent page losing ground should jump the queue.

5. Track outcomes after every refresh (build real ground truth)

Log which pages were refreshed and their rank/CTR changes at 30, 60, and 90 days. This replaces the proxy label with real outcomes and will substantially improve future

model performance.

6. Retrain quarterly; pause if grouped AUC stays near 0.5 for two cycles

The current grouped AUC of 0.507 is the retrain trigger threshold. If two consecutive quarterly retrains fail to exceed 0.55 grouped AUC, the feature set needs re-evaluation before the queue is trusted for editorial decisions.

Hard Limits — What We Will Never Automate

-  Do NOT let a model automatically rewrite and publish an article without human eyes.
-  Do NOT use these scores to evaluate or punish writers.
-  Do NOT auto-delete content because it scores "Low Priority."

7. Artifacts the paper embeds

Generate/collect the charts and tables your deployed page will show.

✓ 7. Reproducibility

All code, notebooks, and pipeline scripts are available in the public repository. The entire pipeline can be re-run in Google Colab with no local setup or paid tools.

Notebook Index

Week	Notebook	Purpose
1	w01_research_question	Lane definition and data first look
2	w02_ml_task_framing	Task type decision: clustering vs classification
3	w03_data_contract	Unit of analysis, feature/label split, leakage demo
3	w03_feature_leakage_check	Formal leakage audit
4	w04_baseline_score	Hand-written rule, ranked queue, leakage date check
5	w05_model	Random Forest training, confusion matrix, error analysis
6	w06_validation_audit	Naive vs grouped split, feature importances, claim rewrite
7	w07_action_playbook	Priority queue with reason codes, CSV export
8	capstone	Full paper integration (this notebook)

To Reproduce

```
# 1. Clone the repo
git clone https://github.com/tokihab/FlyRankIntern-ML

# 2. Install dependencies
pip install -r requirements.txt

# 3. Run the full pipeline
python scripts/run_all.py
```

Or open any Colab badge in the README — no local setup needed.

Full-release data (79M rows) requires requesting access to [FlyRank/internship-warehouse](#) on Hugging Face. Approval is instant after accepting the data-use terms.

Built on the **FlyRank ML Internship dataset** — a production search intelligence dataset released for educational use by [FlyRank](#).

Crediting your data source is standard research practice, and this dataset is the foundation that makes the scale and authenticity of this project possible.

Track leads: Mirza Aščerić (ML) · Hole (Data Engineering)

Code license: MIT (see [LICENSE](#) in repository)

Data license: [DATA_USE.md](#) terms in repository

Repository: <https://github.com/tokihab/FlyRankIntern-ML> Data: <https://flyrank.ai>

Self-check

Before you submit, confirm each line honestly:

- ☐ Every section above is filled — markdown thinking AND the code that backs it
- ☐ The notebook runs top to bottom with no errors (Runtime → Run all)
- ☐ No client names, URLs, or private queries anywhere
- ☐ My claims use careful words: observed, measured, directional, decision-support
- ☐ Committed to my repo under [work/notebooks/](#) — then submit your repo URL on the card. Done.
- ☐ My deployed paper has **all 9 sections** — including the **Abstract** at the top and **Acknowledgments & data credit** (the <https://flyrank.ai> link) at the bottom.

- ☐ **ML-12 done in this notebook's closing cells:** 5-minute demo outline + a social-post cut + a 3-sentence employer-facing summary.